



An EKF observer to estimate semi-autogenous grinding mill hold-ups[☆]



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ARTICLE INFO

Article history:

Received 21 June 2016

Received in revised form 5 December 2016

Accepted 17 December 2016

Available online 9 January 2017

Keywords:

Comminution

Extended Kalman filter

Grinding mill

Observability

Process control

State and parameter estimation

ABSTRACT

A non-linear observer model of a semi-autogenous grinding mill is developed. The observer model distinguishes between the volumetric hold-up of water, solids, and the grinding media in the mill. Solids refer to all ore small enough to discharge through the end-discharge grate, and grinding media refers to the rocks and steel balls. The rocks are all ore too large to discharge from the mill. The observer model uses the accumulation rate of solids and the mill's discharge rate as parameters. It is shown that with mill discharge flow-rate, discharge density, and volumetric hold-up measurements, the model states and parameters are linearly observable. Although instrumentation at the mill discharge is not yet included in industrial circuits because of space restrictions, this study motivates the benefits to be gained from including such instrumentation. An extended Kalman filter is applied in simulation to estimate the model states and parameters from data generated by a semi-autogenous mill simulation model from literature. Results indicate that if sufficiently accurate measurements are available, especially at the discharge of the mill, it is possible to reliably estimate grinding media, solids and water hold-ups within the mill. Such an observer can be used as part of an advanced process control strategy.

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1. Introduction

Process industries in general have benefited considerably from advanced process control, but the mineral processing industry in particular has yet to take full advantage of model-based advanced process control [2]. Simulation studies confirm the considerable benefits advanced model-based process control can achieve over proportional-integral-derivative (PID) control in terms of plant stability and economic performance [3–5]. One of the main impediments to implementing model-based control in the mineral processing industry, and specifically grinding mill circuits, is the lack of sufficient real-time measurements to estimate the necessary model states and parameters. The number of available real-time measurements are generally far less than the size of the state vector to be measured [6]. Therefore, the peripheral tools of the control loop such as observers and soft sensors become as important as the controller itself [7].

In general terms, the grinding performance of a mill depends on the hold-up of steel balls, ore and water. Although the total mill hold-up is generally measured in industrial circuits [6], the steel ball, ore, and water hold-ups cannot be measured directly with the currently available instrumentation. Measurement of these three mill inventories, which influence the mill's product throughput and product quality, can significantly improve plant performance if used to inform a control loop.

The model structure used to describe a comminution process has a significant impact on the success of state and parameter estimation. Models aimed at plant design are generally not suited for model-based control, and vice-versa. Steady-state phenomenological models, a combination between theoretical and empirical models, are well established and have proven valuable for comminution plant design and steady-state optimisation [8–10]. The model parameters are generally divided into the ore to be processed and the comminution unit processing the ore. Laboratory drop-weight and pendulum-breakage tests provide the ore specific parameters, and sampling campaigns provide the processing unit's parameters [9]. Since the characteristics of run-of-mine ore fed to industrial mills change over time, results from laboratory tests used to estimate ore specific parameters do not necessarily reflect the range of ore breakage conditions in industrial

[☆] A subset of this work was presented at the 17th IFAC Symposium on Control, Optimisation and Automation in Mining, Mineral and Metal Processing (Vienna, 2016) [1].

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mills [11]. Also, sampling campaigns used to estimate the processing unit's parameters assume steady-state operation at a specific operating point, but steady-state is difficult to guarantee over a long period given the variation of run-of-mine ore characteristics. It is usually assumed the functions describing the milling environment can be expressed as simple equations with parameters that can be estimated from sampling campaign data through back-calculation. Because of the heavy reliance on back-calculation, small measurement errors can lead to large variances in parameters [12]. With the advancements in computing power, computational fluid dynamics (CFD) and discrete element method (DEM) models have provided valuable insight into the charge motion of grinding processes. A difficulty associated with these fundamental models is the correct estimation of the model's parameters as ore characteristics vary over time [13,14]. Additionally, fundamental models remain too computationally intensive to be used in advanced process control strategies [15,16].

The mathematical models of comminution processes proving most useful for industrial automatic control so far are developed on-line [17]. These empirically derived linear time-invariant transfer function models have been successfully applied in model-based controller strategies to industrial circuits, e.g. by Hulbert et al. [18]; Craig and MacLeod [19]; Chen et al. [20,21]. However, these linear models are restricted to the domain around the nominal operating points of the plant and require constant management to accommodate variations in grinding conditions. System identification is particularly difficult given the inherent uncontrollable disturbances, measurement noise and high tonnage operation [7]. Ideally, real-time measurements should be used to estimate model states and parameters.

Considerable work has been done to estimate grinding mill process variables using different modelling approaches [22]. A very simple model along with power and bearing pressure measurements is used by Herbst et al. [23] to estimate mill filling and rock hardness. This work was extended in Herbst and Pate [24] to estimate ore, water, and ball inventories. A commercialised soft-sensor is described by Herbst and Pate [25] to estimate mill inventories and breakage rates. In all cases a Kalman filter is used to estimate the unknown state vector. Although the filters capture the qualitative trend of the unknown state vector, the studies above do not explicitly include observability analyses to ensure the filters produce reliable solutions.

A linear observability test is included in the inferential measurement work of Apelt et al. [26]. The SAG mill model of Napier-Munn et al. [9] was used to describe the grinding process, along with a novel ball charge model and a mill-liner model. Using measurements of the mill charge weight and size-by-size solids discharge, Apelt et al. [26] used 29 measurements to estimate 37 states and 7 parameters with an extended Kalman filter (EKF). However, the rank of the observability matrix of the linearised system was only 20, which meant a unique solution for the parameters and states was not available.

As shown by Le Roux and Craig [27], it is theoretically possible to uniquely fit the simplified non-linear grinding mill circuit model of Le Roux et al. [28] to real-time plant measurements. However, the algebraic fitting procedure is too sensitive to uncertainties in measurements as the procedure involves the calculation of first and second order time-derivatives from noisy data. Although a relatively simple model is used, there are so many parameters to be defined that the value of this procedure is limited.

The model of Coetzee et al. [29] is used by Olivier et al. [30] to estimate the mill inventories and grinding environment parameters in simulation. Assuming measurements of the mill discharge are available, a dual particle filter is used to estimate the hold-up of balls, rocks, solids, fines, and water as states, and the fraction of rock entering the circuit and the power needed per tonne of fines produced as parameters. The technique achieves good quantitative estimation if constant rock abrasion rates, ball abrasion rates, and step disturbances are assumed, but these assumptions may not necessarily hold true for industrial operation.

Since the grinding performance of a mill depends on the hold-up of steel balls, ore, and water, control of these three hold-ups can reduce the variance in the throughput and quality of the mill's product. Therefore, this work aims at developing an observer for use in model-based control strategies of grinding mill circuits such that the hold-ups in the mill can be regulated [31,32]. To achieve this aim, a non-linear observer model is presented which distinguishes between the volumetric hold-up of water, solids, rocks, and balls in a SAG mill. As seen from an observability analysis, the states and parameters of this model are locally (weakly) observable, but they are not observable from the linearised observer model. The observer model is subsequently reduced such that the states and parameters are linearly observable. The measurements used by the reduced observer model are the total volumetric filling, the mill discharge flow-rate, the mill discharge density, and the derivative of the mill discharge density. Although instrumentation at the mill discharge is not yet included in industrial circuits because of space restrictions, this work motivates the benefits to be gained from including such instrumentation. To illustrate the effectiveness of the observer, a simulation study using an EKF is presented. Results indicate that the modelled states can be reliably estimated if sufficiently accurate measurements are available.

2. Grinding mills

2.1. Process description

The open circuit SAG mill depicted in Fig. 1 receives three streams: mined ore (*MFO*), water (*MIW*) and additional steel balls (*MFB*) to assist with the breakage of ore. If the mill circuit is closed with a classifier such as a hydrocyclone, the underflow from the hydrocyclone also flows into the mill. The mill charge constitutes a mixture of grinding media and slurry. Grinding media refers to the steel balls and large rocks used for breaking the ore, and slurry refers to the mixture of water and all ore material that exhibit the same flow characteristics as water. The fraction of the mill volume filled with charge is denoted by J_f .

The mill is rotated along its longitudinal axis by a motor. As shown in Fig. 2, the charge in the mill is lifted by the inner liners on the walls of the mill to a certain height from where it cascades down, only to be lifted again by the liners through the rotating action of the mill. If the rotational speed is sufficiently fast the material in the charge will become airborne after reaching the top of its travel on the mill shell. The uppermost point where material leaves the mill shell is called the shoulder of the charge. The airborne particles follow a parabolic path, reaching a maximum called the head and making contact again with the mill charge at the bottom of the mill. The cascading motion of the charge causes the ore to break through impact breakage, abrasion and attrition. The mill grind is the fraction of material in the discharge of the mill below the specification size and indicates the efficiency of the mill to break the ore. The power draw (P_{mill}) of the motor turning the mill is an indication of the kinetic and potential energy imparted to the charge.

As shown in Fig. 2, the grinding media starts at the charge shoulder, continues downwards past the slurry toe, and ends at the charge toe. The slurry in the mill stretches from the slurry shoulder to the slurry toe. As slurry in the mill increases, the slurry toe approaches

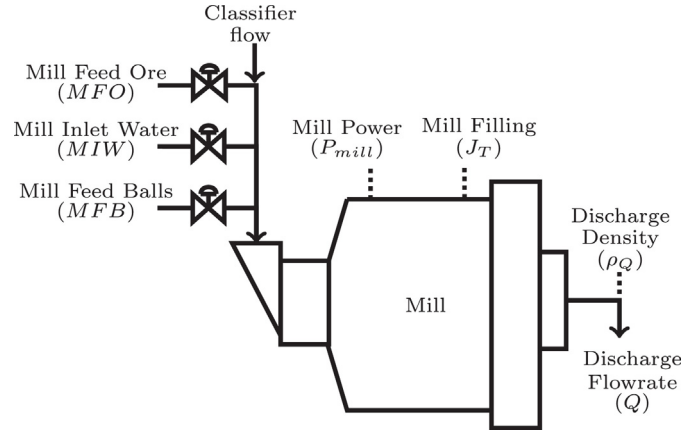


Fig. 1. A semi-autogenous grinding mill.

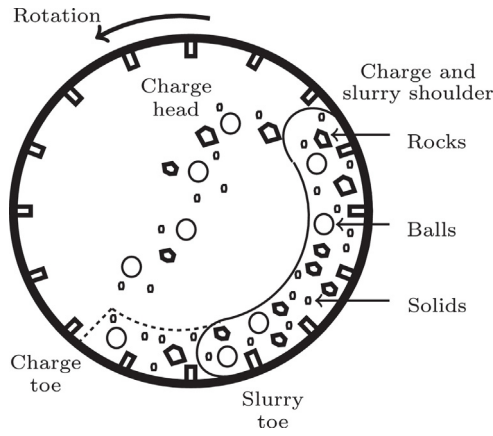


Fig. 2. Cross-section of a semi-autogenous grinding mill.

Table 1
Description of circuit variables assumed to be measured.

Variable	Unit	Description
MIW	m^3/h	Flow-rate of water to the mill
MFO	t/h	Feed-rate of ore to the mill
MFB	t/h	Feed-rate of steel balls to the mill
J_T	–	Fraction of mill volume filled with charge
Q	m^3/h	Mill discharge flow-rate
ρ_Q	t/m^3	Mill discharge density
P_{mill}	kW	Mill power draw

the charge toe. If the slurry toe and charge toe are equal, the slurry completely fills the grinding media. Up to this point, slurry discharge occurs only through the grinding media, i.e. the slurry flows through the grinding media interstices before exiting the mill. If the slurry toe exceeds the charge toe, i.e. there is more slurry than grinding media, a slurry pool forms. Discharge is no longer only via the grinding media, but also occurs via a slurry pool. Slurry pool conditions are undesirable since the pool absorbs the impact of falling material.

The ground ore in the mill mixes with the water to create a slurry. The slurry in a mill begins to form at the shoulder of the charge. The toe of the slurry starts to grow downwards towards the toe of the charge with increasing flow-rate of slurry through the mill. While the toe of the slurry is less than or equal to the toe of the charge, discharge occurs via the grinding media. When the toe of the slurry exceeds the toe of the charge, a slurry pool forms at the bottom of the mill. Slurry discharge is then a combination of flow via the grinding media and the slurry pool [33]. Slurry pool conditions should be avoided as they decrease the mill power draw and the breakage rate by cushioning material falling from the charge shoulder to the charge toe. The slurry is discharged through an end-discharge grate where the aperture size of the end-discharge grate limits the particle size of the discharged slurry. The flow-rate of slurry at the mill discharge is given by Q . It is assumed that the in-mill slurry density is equal to the discharge slurry density (ρ_Q). The circuit variables are described in Table 1.

2.2. Industrial circuit measurements

The survey of Wei and Craig [6] indicates that P_{mill} and J_T are commonly measured variables in industrial plants, whereas Q and ρ_Q are not explicitly included as real-time measured variables for any of the plants surveyed.

Table 2
Observer model nomenclature

Parm	Unit	Description
α_r	–	Mass fraction of rocks in the feed ore
ρ_b	t/m ³	Density of steel balls
ρ_o	t/m ³	Density of feed ore
ρ_w	t/m ³	Density of water
η	h ⁻¹ m ⁻³	Discharge rate per volume of slurry
K_b	1/h	Ball abrasion rate
K_r	1/h	Rock abrasion rate
V_{mill}	m ³	Mill volume
x_b	m ³	Volume of balls in the mill
x_r	m ³	Volume of rocks in the mill
x_s	m ³	Volume of solids in the mill
x_w	m ³	Volume of water in the mill

The combined mass of the mill and of the charge inside the mill is generally measured using either load cells or bearing pressure. Because this is not a direct measurement of J_T , the relation between J_T and a mass measurement needs to be determined whenever a mill survey is performed [34]. For different mill charges, accurate mill filling measurements after mill stops can be used to calibrate the relationship between the mass measurement and J_T . The calibration exercise should be repeated at reasonable intervals as the loss of liner mass through wear and tear will cause a drift in the accuracy of the relationship. Once the drift in the data is quantified, an empirical liner wear model can be constructed to predict the service life of liners and adjust the relationship between the mass measurement and J_T over time. With careful planning, the mass to J_T relationship can easily be checked within half an hour from mill stop to start [35].

This study assumes measurements of Q and ρ_Q are available. Because of space restrictions at the discharge trommel of the mill, inclusion of flow and density instrumentation at the mill discharge is not yet a viable reality [9]. Through careful planning and design of greenfield comminution circuits it should be possible to install existing flow and density instrumentation at a mill discharge trommel. In the case where the mill discharges into a sump, ρ_Q and Q can be back-calculated from a flow-balance if all other inflows and outflows are measured, but this is sensitive to the accuracy of measurements at the sump. This study aims to illustrate the benefits to be gained from including Q and ρ_Q measurement instrumentation in industrial circuits.

SAG mills are usually designed with a constant target fractional volumetric filling of balls (J_B) in mind. Because accurate real-time measurements of J_B are not available, J_B is usually not included in control schemes to manipulate the mill grind and the discharge rate. Although J_B is not measured directly, it can be approximated inferentially using models and measurements of P_{mill} or J_T [36]. Generally a linear relationship between the rock volume and ball abrasion rate is assumed. This assumption allows for the calculation of MFB to maintain an approximately constant J_B in terms of the tonne of ore milled. At steady-state this relates to MFB being a constant fraction of MFO .

3. Observer model for a grinding mill

This section describes how the observer model is developed. The aim of the observer is to use the variables listed in Table 1 as measured quantities to estimate the mill states, the discharge rate constant, and the ore and ball breakage rates. The nomenclature for the observer model is given in Table 2.

The constituents in the mill are modelled as four volumetric quantities: water (x_w), solids (x_s), rocks (x_r), and balls (x_b). The model makes use of only two size classes to describe ore in the mill: *solids* includes all ore smaller than the end-discharge grate aperture size, and *rocks* all ore larger than the aperture size.

3.1. Process dynamics

3.1.1. Process feed

The mill inflow flow-rates are described as

$$V_{wi} = MIW + V_{cw} \quad (1a)$$

$$V_{si} = MFO \frac{1 - \alpha_r}{\rho_o} + V_{cs} \quad (1b)$$

$$V_{ri} = \frac{\alpha_r MFO}{\rho_o} \quad (1c)$$

$$V_{bi} = \frac{MFB}{\rho_b}, \quad (1d)$$

where V_{wi} , V_{si} , V_{ri} , and V_{bi} (m³/h) are the mill inflow of water, solids, rocks, and balls respectively, ρ_o (t/m³) is the density of the feed ore, ρ_b (t/m³) is the density of the steel balls, and V_{cw} and V_{cs} (m³/h) are the flow of water and solids returned by the classifier, respectively. The parameter α_r defines the mass fraction of rocks in MFO and is assumed to be measured as a function of time [6]. The implicit assumption when dividing by ρ_o in (1b) and (1c) is that the ore is non-porous. It is assumed that the flow from the classifier to the mill is known.

3.1.2. Population balance

The population balance used to describe the kinetics of the states defined above is

$$\dot{x}_w = V_{wi} - V_{wo} \quad (2a)$$

$$\dot{x}_s = V_{si} - V_{so} + RC \quad (2b)$$

$$\dot{x}_r = V_{ri} - RC \quad (2c)$$

$$\dot{x}_b = V_{bi} - BC, \quad (2d)$$

where V_{wo} and V_{so} (m^3/h) are the discharge of water and solids from the mill respectively, and RC and BC (m^3/h) are the consumption of rocks and balls respectively. Because the mill is fitted with an end-discharge grate, no rocks or balls are discharged from the mill. It is assumed the mill is a fully mixed reactor. The rate of ball consumption is considerably lower than for rocks, and its contribution to $\dot{x}_s$ is regarded as negligible.

3.1.3. Breakage rates

The cumulative rates modelling approach assumes only one function is necessary to describe the grinding kinetics. The cumulative breakage rate function is defined as the rate per unit mass that a given species coarser than a given size breaks to below that size [37]. This is an advantage over other population balance models which require two functions, a breakage rate and an appearance function, to describe grinding kinetics [38]. The cumulative breakage rates modelling approach was used by Austin et al. [39] to model the steady-state behaviour and by Amestica et al. [40,41] to model the dynamic behaviour of SAG mills. A disadvantage of this approach is that the cumulative rate of breakage of ore above a given size s_i is assumed to be unaffected by the grinding environment and the detailed structure of the size distribution above s_i . Empirical relationships are required to relate the parameters of the cumulative breakage rate function to variations in the milling environment. However, more sophisticated models face similar problems where the parameters of both the breakage rate and breakage distribution function vary according to changes in the grinding environment [12]. The cumulative breakage rate function remains constant if the ball filling level and the charge level remain relatively constant [40]. A validation study of the cumulative rates model was conducted by Salazar et al. [42], and a model predictive controller using the cumulative rates model was investigated in simulation by Salazar et al. [43].

Similar to the cumulative rates modelling approach, the consumption of rocks (RC) and balls (BC) in the population balance of (2b) to (2d) are described as

$$RC = x_r K_r \quad (3a)$$

$$BC = x_b K_b, \quad (3b)$$

where K_r and K_b ($1/\text{h}$) are the abrasion rates of rocks and balls, respectively. Instead of the first order approach above, a second order model could be used to describe RC as the interaction between rocks and rocks, balls and rocks, and rocks and liners. However, this is deemed as an unnecessary increase in the number of model parameters as the first order cumulative rates modelling approach is well attested in literature and practice [40,41,12,42,43].

A relatively constant rock abrasion rate K_r can be achieved through a high x_b (as in the case of a ball mill), but the heavy balls increase the power required to turn the mill and consequently increase the energy costs. The ball abrasion rate K_b depends on the ore characteristics, the mill liner type, the ball material, x_r , and x_b . A high x_b increases K_b as there is more ball-ball and ball-liner contact rather than ball-ore contact. Although a low x_b reduces K_b , it also reduces the grinding ability of the mill. For a very high mill rotational rate the balls may collide with exposed liners causing unnecessary liner wear and a higher K_b .

3.1.4. Mill discharge

The mill discharge model for the observer is adapted from the model presented in Morrell and Stephenson [44], where the discharge of charge from the mill through the end-discharge grate is viewed as a product of two mechanisms: the fluid transport through the grate, and the solids classification by the grate which differentiates between ground particles either returning to the milling chamber for further grinding or becoming part of the mill discharge stream. In the case of $i = 1, 2, \dots, N$ ore size classes, the discharge rate (d_i) (h^{-1}) of the mill charge is approximated as [45]

$$\begin{aligned} d_i &= d_0; & s_i &\leq s_m \\ d_i &= d_0 \frac{\ln s_i - \ln s_g}{\ln s_m - \ln s_g}; & s_m &< s_i \leq s_g \\ d_i &= 0; & s_i &> s_g, \end{aligned} \quad (4)$$

where s_i (mm) represents the particle size classes, d_0 (h^{-1}) is the specific discharge rate for water and ore up to particle size s_m (mm) (i.e. slurry), and s_g (mm) is the effective mesh size of the grate above which discharge is zero. Since rocks and balls are larger than s_g , they have a discharge rate of 0/h.

To determine d_0 , a function relating slurry hold-up to slurry flow (i.e. flow of particles smaller than s_m exhibiting the same flow characteristics as water) out of the mill is required. This function should account for whether the slurry flow occurs via the grinding media or via a slurry pool. The slurry flowing through a single grate aperture depends on the following factors: area of the aperture, depth of the aperture below the free surface of the slurry, gravity and centrifugal forces, and slurry viscosity. By analysing pilot and industrial mills, Morrell and Stephenson [44] expressed the relationship between slurry flow-rate and hold-up for the case where slurry flow occurs via the grinding media and for the case where slurry flow occurs via the slurry pool.

As discussed in Section 2.1, slurry flow via the grinding media occurs when $J_p \leq J_{max}$ and via the slurry pool when $J_p > J_{max}$. Here J_p is the fraction of the mill volume filled with slurry, $J_{max} = 0.5J_T - J_{po}$ is the maximum volumetric fraction of the interstices of the grinding media which can be filled with slurry, $J_{po} = 0.33(1 - \bar{r}_a)$ is the fraction of “dead” slurry hold-up within the mill (contained between the mill shell and the outer most grate aperture), and $\bar{r}_a$ (m) is the radial position of the outermost row of grate apertures as a fraction of the mill radius r_m .

Slurry flow via the grinding media Q_m (m³/h) is expressed as [44]

$$Q_m = k_m J_{pm}^2 \lambda^{2.5} D_m^{0.5} A \phi_c^{-1.38} \quad (5)$$

where k_m (m^{0.5}/h) is a constant, J_{pm} is the net fractional slurry hold-up in the grinding media, D_m (m) is the mill diameter, A (m²) is the total open area of the grate apertures, ϕ_c is the fraction of the critical mill speed. The mean relative radial position of the grate apertures is defined as $\lambda = (\sum r_i a_i) / (r_m \sum a_i)$ where a_i is the open area of all holes at a radial position r_i . Slurry flow via the slurry pool Q_t (m³/h) is expressed as

$$Q_t = k_t J_{pt} \lambda^2 D_m^{0.5} A \quad (6)$$

where $J_{pt} = J_p - J_{max}$ is the net fractional hold-up of slurry in the slurry pool, and k_t (m^{0.5}/h) is a constant.

It is worth noting the inverse relationship between ϕ_c and Q_m shown in (5). A possible reason for this relationship is the increase in the dynamic porosity (void fraction) of the charge as the mill speed increases [33]. Also, the grinding media thrown from the charge shoulder overshoots the charge toe because of the higher speed. As a result of the overshoot the number of high impact breakage events reduces and consequently the fines contributing to the mill slurry reduce. Morrell and Stephenson [44] assumed the viscosity effects would be captured by the constants k_m and k_t without altering the form of (5) and (6).

Solids between sizes s_m and s_g also exit the mill and contribute approximately 5–15% to the volumetric flow-rate depending on the size of the grates. To predict the total flow-rate of slurry out of the mill Q , the predictions for Q_m and Q_t need to be increased to account for the additional amount of coarse material. Therefore

$$Q = k_g (Q_m + Q_t), \quad (7)$$

where k_g is a positive dimensionless factor to account for coarse material. The value of k_g varies depending on the grate aperture size. Morrell and Stephenson [44] provide guidelines on the choice of k_g .

In the inferential measurement work of Apelt et al. [26], it is assumed that no slurry pooling occurs in the case of a large open area (A), a high relative radial position of the open area (λ), and a high relative radial position of the outermost apertures ($\bar{r}_a$) in the end-discharge grate. For this study it was assumed that the discharge flow is only through the grinding media, i.e. $Q = k_g Q_m$. Therefore, assuming no slurry pooling, Q can be approximated as [46]

$$Q = k_g k_m \lambda^{2.5} D_m^{0.5} A \phi_c^{-1.38} J_{pm}^2. \quad (8)$$

As shown in (8), Q is quadratically proportional to J_{pm} if there is no slurry pool. Thus, in terms of the four states of the observer defined (x_w , x_s , x_r , and x_b), it is possible to express Q as

$$Q = \eta (x_w + x_s)^2 \quad (9)$$

where $(x_w + x_s)$ represents the total slurry hold-up in the mill, and η (h⁻¹m⁻³) is the discharge rate per volume of slurry. Since only two size classes are used for the observer model, one size which discharges completely and one which remains inside the mill, it is implicitly assumed in (9) that $s_g = s_m$ in (4). Therefore, the discharge of the water (V_{wo}) and solids (V_{so}) in (2a) and (2b) can be expressed as

$$V_{wo} = \eta (x_w + x_s) x_w \quad (10a)$$

$$V_{so} = \eta (x_w + x_s) x_s. \quad (10b)$$

3.2. Process output

For this study, it is assumed that measurements of J_T , Q and ρ_Q are available. The implications of measuring these variables are discussed in Section 2.2. These quantities are modelled as

$$J_T = \frac{x_w + x_s + x_r + x_b}{v_{mill}} \quad (11a)$$

$$Q = \eta (x_w + x_s)^2 \quad (11b)$$

$$\rho_Q = \frac{\rho_o x_s + \rho_w x_w}{x_s + x_w}. \quad (11c)$$

The fourth measured output listed in Table 1 is P_{mill} (kW), which is defined by Apelt et al. [36] as

$$P_{mill} = P_{NL} + k_p P_C, \quad (12)$$

where P_{NL} (kW) is the no-load power (empty mill power draw), P_C (kW) is the mill power draw attributed to the mill contents, and k_p is a lumped power draw parameter to account for heat losses due to internal friction, energy for attrition and abrasion breakage and rotation of grinding media, plus inaccuracies associated with the assumptions and measurements of the charge shape and motion [9].

Apelt et al. [36] models P_C as an empirical function of J_T and the mill charge density (ρ_C)

$$\rho_C = \frac{\rho_b x_b + \rho_o (x_r + x_s) + \rho_w x_w}{x_b + x_r + x_s + x_w}. \quad (13)$$

Ideally ρ_C could be inferred from the measurement of P_{mill} using the power draw model in Apelt et al. [36]. However, to use the model, parameter k_p in (12) must be fitted to process data. This fitting process requires ρ_C to be known. Therefore, the observer model does not make use of P_{mill} as an output equation as it only introduces additional parameters to estimate. The same holds true for other models of P_{mill} , such as for example the models in Austin [47] and Le Roux et al. [28].

4. Observability of states and parameters

4.1. Background

A multi-input-multi-output control-affine non-linear state-space model with $\dim(x) = n$ and $\dim(y) = m$ can be written as

$$\begin{aligned}\dot{x} &= f(x) + g(x)u \\ y &= h(x).\end{aligned}\quad (14)$$

The system in (14) is said to be locally (weakly) observable at x_0 if there exists a neighbourhood X_0 of x_0 such that for every x_1 which is an element of the neighbourhood $X_1 \subset X_0$ of x_0 the indistinguishability of the states x_0 and x_1 implies that $x_0 = x_1$. The two states x_1 and x_0 are said to be indistinguishable if for every admissible input u the output y of (14) for the initial state x_0 and for the initial state x_1 is identical. If the system satisfies the so called observability rank condition, i.e. the observability codistribution of x_0 [48]

$$d\mathcal{O} = \text{span} \left\{ dh_j, dL_f h_j, \dots, dL_f^{n-1} h_j \right\}; \quad j = 1, \dots, m \quad (15)$$

has dimension n at x_0 , then the system is locally (weakly) observable. Note, $L_f^k h_j$ refers to the k th repeated Lie derivative of the scalar function $h_j(x)$ along the vector field $f(x)$, and d is the exterior derivative. In the linear case, the observability codistribution corresponds to the observability matrix $\mathcal{O}^T = [C^T, A^T C^T, \dots, (A^{n-1})^T C^T]$ where $C = \frac{\partial h}{\partial x}|_{x=x_0}$ and $A = \frac{\partial}{\partial x}(f(x) + g(x)u)|_{x=x_0}, u=u_0$.

4.2. Analysis of observer model

For the observer model of the grinding mill, it is assumed that the parameters η, K_r and K_b are unknown constants, although in practice these parameters may vary slowly. Thus, the observer model described in Section 3 can be written in the form of (14), such that

$$\dot{x} = f_O(x) + g_O(x)u = \begin{bmatrix} -\eta(x_w + x_s)x_w \\ -\eta(x_w + x_s)x_s + x_r K_r \\ -x_r K_r \\ -x_b K_b \\ 0_{3 \times 1} \end{bmatrix} + \begin{bmatrix} I_{4 \times 4} \\ 0_{3 \times 4} \end{bmatrix} u \quad (16a)$$

$$y = h_O(x) = [J_T, Q, \rho_Q]^T \quad (16b)$$

where $x = [x_w, x_s, x_r, x_b, \eta, K_r, K_b]^T$, $u = [V_{wi}, V_{si}, V_{ri}, V_{bi}]^T$ and $y = h_O(x)$ is given by (11). (The subscript O refers to *Observer*.)

For the system in (16), the dimension of the observability codistribution $d\mathcal{O}$ was determined using Maple.¹ The dimension of $d\mathcal{O}$ at a generic point x_0 is 7, which implies that the non-linear system is locally (weakly) observable. Thus, a non-linear observer such as a moving horizon estimator (MHE) could possibly estimate the unknown states and parameters. However, it is necessary to correctly assume the time-varying nature of the parameters for the MHE to estimate the true state and parameter values. A longer time horizon for the MHE reduces the validity of modelling the parameters as constants, but a shorter time horizon may not include sufficient system dynamics for the observer to estimate the unknown states and parameters.

Linearisation of (16) results in the A and C matrices

$$A = \begin{bmatrix} -\eta(2x_w + x_s) & -\eta x_w & 0 & 0 & -(x_w + x_s)x_w & 0 & 0 \\ -\eta x_s & -\eta(x_w + 2x_s) & K_r & 0 & -(x_w + x_s)x_s & x_r & 0 \\ 0 & 0 & -K_r & 0 & 0 & -x_r & 0 \\ 0 & 0 & 0 & -K_b & 0 & 0 & -x_b \\ 0_{3 \times 1} & 0_{3 \times 1} & 0_{3 \times 1} & 0_{3 \times 1} & 0_{3 \times 1} & 0_{3 \times 1} & 0_{3 \times 1} \end{bmatrix} \quad (17a)$$

$$C = \begin{bmatrix} \frac{1}{v_{mill}} & \frac{1}{v_{mill}} & \frac{1}{v_{mill}} & \frac{1}{v_{mill}} & 0 & 0_{1 \times 2} \\ 2\eta(x_w + x_s) & 2\eta(x_w + x_s) & 0 & 0 & (x_w + x_s)^2 & 0_{1 \times 2} \\ \frac{\rho_w - \rho_Q}{x_w + x_s} & \frac{\rho_o - \rho_Q}{x_w + x_s} & 0 & 0 & 0 & 0_{1 \times 2} \end{bmatrix}. \quad (17b)$$

The observability matrix $\mathcal{O}$ formed from (17) has rank 5. From an analysis of the eigenvalues and eigenvectors of the matrix A it can be seen that the three unknown parameters (η, K_r, K_b) each contribute an integral mode to the linearised system. Because of this repeated mode, two of the three parameters need to be chosen in the linearised model for the other parameter to be linearly observable.

In an attempt to attain linear observability of the system, the time derivatives of (16b) can be used as additional measurements. However, the addition of $\dot{J}_T, \dot{Q}$, or $\dot{\rho}_Q$, or all of these derivatives only increases the rank of the observability matrix $\mathcal{O}$ from 5 to 6. To achieve full column rank, the measurement ρ_C needs to be used along with $\dot{Q}$, or $\dot{\rho}_Q$, or both. However, direct measurement of ρ_C is rarely possible. If

¹ MapleTM is a trademark of Waterloo Maple Inc.

ρ_C is estimated from P_{mill} , additional unknown parameters are introduced which negates any benefit of using ρ_C as an inferred measured variable.

4.3. Model reduction and linear observability

4.3.1. Reduced model

To achieve linear observability, the model in (16) is reduced based on the assumption that the dynamics of the slurry (x_w and x_s) is much faster than the dynamics of the grinding media (x_r and x_b). Therefore,

$$\dot{x} = f_{RM}(x) + g_{RM}(x)u = \begin{bmatrix} -\eta(x_w + x_s)x_w \\ -\eta(x_w + x_s)x_s + \chi \\ 0_{3 \times 1} \end{bmatrix} + \begin{bmatrix} I_{2 \times 2} \\ 0_{3 \times 2} \end{bmatrix} u \quad (18a)$$

$$y = h_{RM}(x, u) = \begin{bmatrix} \frac{x_w + x_s + x_{rb}}{v_{mill}} \\ \eta(x_w + x_s)^2 \\ \frac{\rho_o x_s + \rho_w x_w}{x_s + x_w} \\ \frac{(\rho_o - \rho_w)(\chi x_w + x_w V_{si} - x_s V_{wi})}{(x_s + x_w)^2} \end{bmatrix} \quad (18b)$$

where $x = [x_w, x_s, x_{rb}, \eta, \chi]^T$, $u = [V_{wi}, V_{si}]^T$, and $y = [f_T, Q, \rho_Q, \dot{\rho}_Q]^T$. (The subscript RM refers to *Reduced Model*.) The state x_{rb} (m^3) represents all grinding media in the mill (sum of rocks and balls) which is assumed to stay relatively constant compared to x_w and x_s , and the state χ (m^3/h) represents the generation of solids. The output vector in (11) is extended in (18b) with the addition of $\dot{\rho}_Q$.

4.3.2. Linear observability

The A and C matrices for the system in (18) are

$$A = \begin{bmatrix} -\eta(2x_w + x_s) & -\eta x_w & 0 & -(x_w + x_s)x_w & 0 \\ -\eta x_s & -\eta(x_w + 2x_s) & 0 & -(x_w + x_s)x_s & 1 \\ 0_{3 \times 1} & 0_{3 \times 1} & 0_{3 \times 1} & 0_{3 \times 1} & 0_{3 \times 1} \end{bmatrix} \quad (19a)$$

$$C = \begin{bmatrix} \frac{1}{v_{mill}} & \frac{1}{v_{mill}} & \frac{1}{v_{mill}} & 0 & 0 \\ 2\eta(x_w + x_s) & 2\eta(x_w + x_s) & 0 & (x_w + x_s)^2 & 0 \\ \frac{\rho_w - \rho_Q}{x_w + x_s} & \frac{\rho_o - \rho_Q}{x_w + x_s} & 0 & 0 & 0 \\ \frac{(\rho_o - \rho_w)(\chi + V_{si})}{(x_w + x_s)^2} - \frac{2\dot{\rho}_Q}{x_w + x_s} & \frac{(\rho_w - \rho_o)V_{wi}(x_w + x_s)^2}{-} & \frac{2\dot{\rho}_Q}{x_w + x_s} & 0 & 0 & \frac{(\rho_o - \rho_w)x_w}{(x_w + x_s)^2} \end{bmatrix}. \quad (19b)$$

The observability matrix $\mathcal{O}$ of the pair (C, A) has full column rank. Therefore, all states and parameters are observable from the system outputs. However, at steady-state, $\dot{\rho}_Q$ will be equal to zero and the observability matrix $\mathcal{O}$ will no longer have full column rank. (The determinant of the first 5 rows and columns is $\det(\mathcal{O}_{1..5,1..5}) = (\rho_o - \rho_w)\dot{\rho}_Q/v_{mill}^2$ which reduces to zero at steady-state.) For a reasonable value of $\dot{\rho}_Q$ it is necessary to continuously excite the system.

As seen in (18b), the derivative of ρ_Q is used as a fourth measurement. This is necessary to achieve linear observability. If $\dot{f}_T$ is used instead of $\dot{\rho}_Q$ in (18b), the observability matrix $\mathcal{O}$ does not have full column rank. Although $\dot{Q}$ could be used to achieve full column rank, it is a function of $\dot{\eta}$ which is assumed to be equal to zero in (18a). Rather than including the assumption that $\dot{\eta} = 0$ in the measurement model, $\dot{\rho}_Q$ is used in (18b).

5. Observer design

The trapezoidal rule is used to discretize the reduced observer model presented in continuous-time form in (18). A discrete-time EKF is used as the non-linear estimator [49]. The system and measurement equations are

$$\begin{aligned} x_k &= F_{k-1}(x_{k-1}, u_{k-1}, w_{k-1}) \\ y_k &= H_k(x_k, u_k, v_k) \\ w_k &\sim (0, Q_k); v_k \sim (0, R_k), \end{aligned} \quad (20)$$

where the process noise w_k is white noise with covariance $Q_k > 0$ and the measurement noise v_k is white noise with covariance $R_k > 0$.

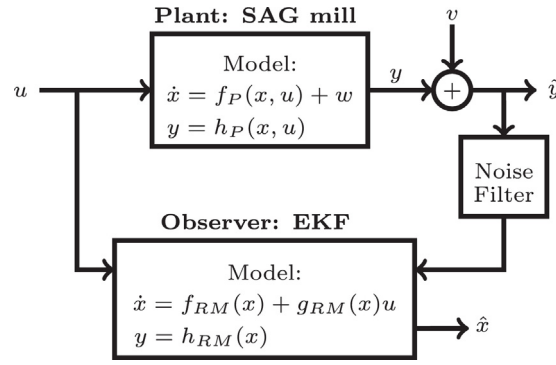


Fig. 3. Simulation setup.

Between each measurement, the state estimate $\hat{x}_k$ and the estimation-error covariance matrix P_k is propagated according to the known non-linear dynamics of the system:

$$\begin{aligned}\hat{x}_k^- &= F_{k-1}(\hat{x}_{k-1}^+, u_{k-1}, 0) \\ P_k^- &= T_{k-1}P_{k-1}^+T_{k-1}^T + L_{k-1}Q_{k-1}L_{k-1}^T\end{aligned}\quad (21)$$

where $T_{k-1} = \frac{\partial F_{k-1}}{\partial x_{k-1}}|_{\hat{x}_{k-1}^+, u_{k-1}, 0}$ and $L_{k-1} = \frac{\partial F_{k-1}}{\partial w_{k-1}}|_{\hat{x}_{k-1}^+, u_{k-1}, 0}$. The state estimate and its covariance is updated through

$$\begin{aligned}K_k &= P_k^-S_k^T(S_kP_k^-S_k^T + R_k)^{-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k[y_k - H_k(\hat{x}_k^-, u_k, 0)] \\ P_k^+ &= (I - K_kS_k)P_k^-\end{aligned}\quad (22)$$

where $S_k = \frac{\partial H_k}{\partial x_k}|_{\hat{x}_k^-, u_k, 0}$.

6. Simulation

6.1. Simulation setup

The aim of the simulation is to test the effectiveness of the reduced observer model to be used in a state and parameter estimation scheme. The simulation setup is depicted in Fig. 3. The plant, a SAG mill with an end-discharge grate, is represented by the continuous-time dynamic non-linear model of Le Roux et al. [28], and the power-draw is given by the model of Apelt et al. [36]. This model is used to simulate the system and to generate fictitious measurement data. The noisy measurement data is filtered using a Savitzky-Golay filter, which is also used to determine the derivative $\dot{\rho}_Q$ [50].

6.1.1. Plant model

The SAG mill simulation model of Le Roux et al. [28] was developed with the aim to produce reasonably accurate model responses using as few parameters and states as possible. The model was validated using industrial plant data. The model divides the ore into the same two size classes as used in the observer model. Therefore, the mill inflow flow-rates are the same as in (1), and the population balance of the model is the same as in (2).

The mill discharge flow-rates are defined as

$$V_{wop} = \varphi d_H x_w \left(\frac{x_w}{x_s + x_w} \right) \quad (23a)$$

$$V_{sop} = \varphi d_H x_w \left(\frac{x_s}{x_s + x_w} \right), \quad (23b)$$

where d_H (1/h) is a constant discharge rate parameter, and φ is an empirical function called the rheology factor. (The subscript P refers to Plant.) The rheology factor attempts to incorporate the effect of the fluidity and density of the slurry on the performance of the milling circuit and is defined as

$$\varphi = \left[\max \left(0, 1 - \left(\frac{1}{\varepsilon_{sv}} - 1 \right) \frac{x_s}{x_w} \right) \right]^{0.5}, \quad (24)$$

where ε_{sv} is the maximum fraction of solids by volume of slurry at zero slurry flow. A rheology factor of unity corresponds to $\frac{x_s}{x_w} = 0$, indicating the slurry consists only of water. A rheology factor of zero corresponds to $\frac{x_s}{x_w} = \frac{\varepsilon_{sv}}{1 - \varepsilon_{sv}}$, indicating the slurry is a non-flowing mud.

The rock consumption (RC_P) and ball consumption (BC_P) for the plant model are defined as

$$RC_P = \frac{\varphi P_{mill} x_r}{\rho_o \kappa_r (x_r + x_s)} \quad (25a)$$

$$BC_P = \frac{\varphi P_{mill} x_b}{\kappa_b [\rho_o (x_r + x_s) + \rho_b x_b]}, \quad (25b)$$

Table 3
Model parameter values.

Parameter	Value	Unit	Description
α_r	0.47	–	Fraction of rock in feed ore
d_H	88	1/h	Constant mill discharge parameter
D_m	4.07	m	Mill inside diameter ($D_m = 2r_m$)
ε_{sv}	0.6	–	Max fraction solids by volume of slurry at zero slurry flow
ϕ_c	0.72	–	Fraction of critical mill speed
g	9.8	m ² /s	Gravity constant
κ_b	90	kWh/t	Steel ball abrasion energy requirement
κ_r	6.72	kWh/t	Rock abrasion energy requirement
k_p	0.97	–	Power draw fitting parameter
L_c	0	m	Mill cone length
L_m	4.54	m	Mill cylinder length
P_{NL}	93.73	kW	Mill power at zero load
ρ_b	7.85	t/m ³	Steel ball density
ρ_o	3.2	t/m ³	Ore density
ρ_w	1	t/m ³	Water density
r_t	0.46	m	Mill feed trunnion radius
V_{cs}	96.9	m ³ /h	Flow of classifier solids to mill
V_{cw}	112	m ³ /h	Flow of classifier water to mill
v_{mill}	59.12	m ³	Mill volume

where κ_r and κ_b (kWh/t) represent the energy requirement for the abrasion of rocks and balls, respectively. The mill power draw model described in Apelt et al. [36] is used to define P_{mill} .

Finally, the plant model is written as

$$\dot{x} = f_p(x, u) = \begin{bmatrix} V_{wi} - \varphi d_H x_w \left(\frac{x_w}{x_s + x_w} \right) \\ V_{si} - \varphi d_H x_w \left(\frac{x_s}{x_s + x_w} \right) + \frac{\varphi P_{mill} x_r}{\rho_o \kappa_r (x_r + x_s)} \\ V_{ri} - \frac{\varphi P_{mill} x_r}{\rho_o \kappa_r (x_r + x_s)} \\ V_{bi} - \frac{\varphi P_{mill} x_b}{\kappa_b [\rho_o (x_r + x_s) + \rho_b x_b]} \end{bmatrix} \quad (26a)$$

$$y = h_p(x, u) = \begin{bmatrix} \frac{x_w + x_s + x_r + x_b}{v_{mill}} \\ \varphi d_H x_w \\ \frac{\rho_o x_s + \rho_w x_w}{x_s + x_w} \end{bmatrix} \quad (26b)$$

where $y = [J_T, Q, \rho_Q]^T$.

6.2. Simulation environment

Table 3 shows the plant model parameter values and Table 4 shows the initial conditions considered. The data is adapted from the industrial data in Le Roux et al. [28] for a single-stage closed grinding mill circuit.

6.2.1. Plant and observer

The general simulation environment for the plant and observer is as follows:

Table 4
Initial operating conditions.

Variable	Value	σ	Unit
<i>Inputs and outputs</i>			
MIW	4.64	–	m ³ /h
MFO	65.2	–	t/h
MFB	5.68	–	t/h
J_T	0.33	0.006	–
Q	234	5	m ³ /h
ρ_Q	2.10	0.02	t/m ³
<i>Plant initial states</i>			
x_b	8.23	0.04	m ³
x_r	1.88	0.01	m ³
x_s	4.65	0.02	m ³
x_w	4.63	0.02	m ³

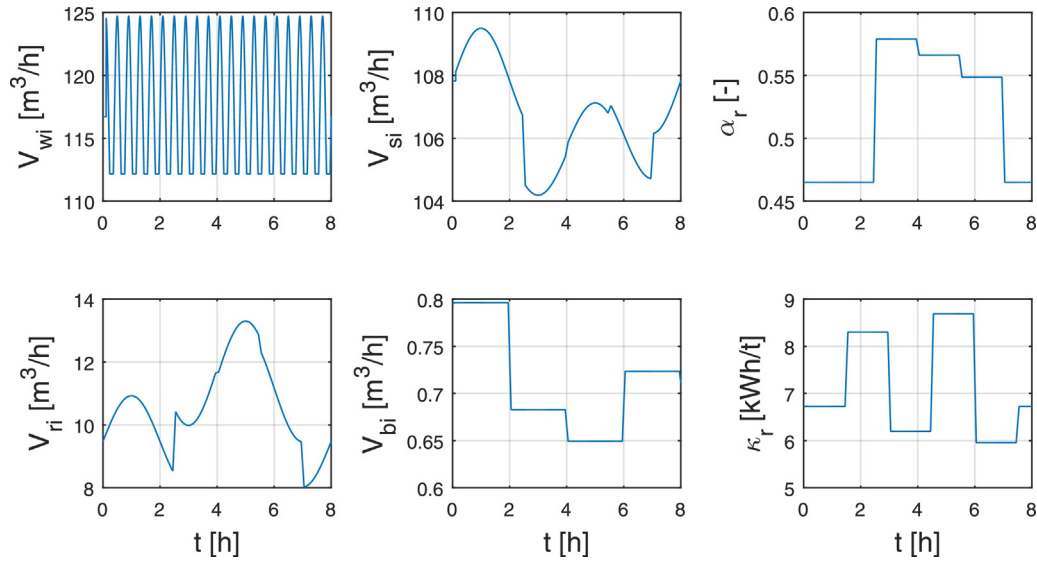


Fig. 4. Inputs as described in (1), and disturbances for simulation scenarios.

- 1 The plant is integrated using the fourth order Runge–Kutta method at a rate of $T_s = 2$ s. The full simulation time is 8 h.
- 2 Measurements are sampled at a rate of $T_s = 2$ s.
- 3 There is no feedback controller for the mill. For the sake of experimentation, it is assumed the flow from the classifier to the mill remains constant throughout the simulation.
- 4 The plant is excited by sinusoidally varying input MIW with a period of $T_{MIW} = 12$ min and an amplitude of $8 \text{ m}^3/\text{h}$ around its initial condition. Since the amplitude is greater than the initial condition, the additional bound $MIW \geq 0 \text{ m}^3/\text{h}$ is considered.
- 5 To show the ability of the observer to track changes in x_s , and x_{rb} , inputs MFO and MFB are varied. Input MFO is sinusoidally varied at a period of $T_{MFO} = 4$ h and an amplitude of 10 t/h . Step changes at intervals of 2 h are applied to MFB where the deviations from the operating point are chosen from the uniform distribution $\mathcal{U}(-1, 1) \text{ (t/h)}$.
- 6 Disturbances in the feed-ore size-distribution and the feed-ore hardness are simulated by varying α_r and κ_r around their nominal values.
 - A change in feed size distribution is simulated by applying step-changes to α_r at intervals of 1.5 h starting at $t = 2.5$ h. The size of step-changes are randomly selected from the uniform distribution $\mathcal{U}(-0.14, 0.14)$. It is assumed measurement instrumentation is available to measure α_r at the input of the plant [6].
 - A change in feed ore hardness is simulated by applying step-changes to κ_r at intervals of 1.5 h starting at $t = 1.5$ h. The size of step-changes are randomly selected from the uniform distribution $\mathcal{U}(-2, 2) \text{ (kWh/t)}$. This parameter is not measured.

Fig. 4 shows the inputs and disturbances to the plant.

6.2.2. Simulation scenarios

Two simulation scenarios are considered.

Simulation 1. No measurement noise or process noise is added. Perfect measurements of the outputs in (26b) as well as the derivative of ρ_Q is assumed and is passed to the observer, i.e. R_k is small. The covariance matrix Q_k is used as a tuning matrix. The initialisation of the observer is:

- $\hat{x}_0 = [3.5 \text{ m}^3, 4 \text{ m}^3, 9 \text{ m}^3, 3 \text{ 1/(hm}^3), 10 \text{ m}^3/\text{h}]^T$
- $P_0 = (\text{diag}[2 \text{ m}^3, 2 \text{ m}^3, 2 \text{ m}^3, 2 \text{ 1/(hm}^3), 2 \text{ m}^3/\text{h}])^2$
- $Q_k = 10^{-1} \times (\text{diag}[0.2 \text{ m}^3, 0.5 \text{ m}^3, 1 \text{ m}^3, 0.8 \text{ 1/(hm}^3), 2 \text{ m}^3/\text{h}])^2$
- $R_k = 10^{-4} \times (\text{diag}[1, 1 \text{ m}^3/\text{h}, 1 \text{ t/m}^3, 1 \text{ t/(hm}^3)])^2$

Simulation 2. Measurement noise (v) is added to the outputs of the plant with a normal distribution of $\mathcal{N}(0, \sigma_v^2)$. System noise (w) is added to the plant states with a normal distribution of $\mathcal{N}(0, \sigma_w^2)$. The values σ_v and σ_w are shown in Table 4.

The covariance matrix R_k for the EKF aligns with the covariance of the measurement noise added to the plant outputs. A high covariance is assigned to the measurement of $\dot{\rho}_Q$ as this measurement is highly sensitive to the noise on ρ_Q . The covariance matrix Q_k is used as a tuning matrix. The initialisation of the observer is:

- $\hat{x}_0 = [3.5 \text{ m}^3, 4 \text{ m}^3, 9 \text{ m}^3, 3 \text{ 1/(hm}^3), 10 \text{ m}^3/\text{h}]^T$
- $P_0 = (\text{diag}[2 \text{ m}^3, 2 \text{ m}^3, 2 \text{ m}^3, 2 \text{ 1/(hm}^3), 2 \text{ m}^3/\text{h}])^2$
- $Q_k = (\text{diag}[0.1 \text{ m}^3, 0.1 \text{ m}^3, 0.1 \text{ m}^3, 0.2 \text{ 1/(hm}^3), 2 \text{ m}^3/\text{h}])^2$
- $R_k = (\text{diag}[0.006, 5 \text{ m}^3/\text{h}, 0.02 \text{ t/m}^3, 1.0 \text{ t/(hm}^3)])^2$

For the second simulation scenario where noise is added, the noisy measurements are filtered using a Savitzky–Golay digital filter. The filter smooths data by fitting a frame of F data samples with polynomials of small order N by means of linear least squares. The filter returns

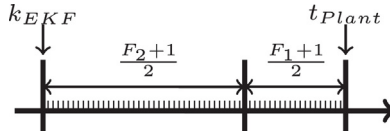


Fig. 5. Filter implementation and management of phase shift.

a filtered value at the centre of the frame. For real-time applications this means that the value returned by the filter is delayed by half the frame length $((F+1)/2)$. To minimise this time delay, two consecutive filters are used.

- (1) The three measurements described in (26b) are filtered with a filter configured to fit a polynomial of order 2 using a frame length of $F_1 = 35$ samples.
- (2) The first order derivative $\dot{\rho}_Q$ is calculated from the delayed filtered ρ_Q data. A Savitzky-Golay filter with an order of 2 and a frame length of $F_2 = 135$ is used.

As depicted in Fig. 5, the filtered measurements are delayed by 18 samples from the current sample. Because $\dot{\rho}_Q$ is calculated from the filtered data, another 68 samples delay is introduced. The equivalent delay is 86 samples, i.e. 172 s.

7. Results and discussion

7.1. Simulation Scenario 1

The results of the simulation scenario where perfect measurements and no process noise are assumed is shown in Figs. 6 and 7. As seen from Fig. 7, the filter converges within two hours to the correct state value and is able to track the changes in the states. The effect of the disturbances in parameter κ_r given in Fig. 4, which affects the generation of solids, is clearly visible in the abrupt changes of χ in Fig. 7, which represents the generation of solids.

7.2. Simulation Scenario 2

The results of the simulation scenario where measurements are filtered to remove noise and to calculate the derivative of ρ_Q is shown in Figs. 8 and 9. As seen in Fig. 8, the noise on the measurements of J_T , Q and ρ_Q are filtered very well, but the derivative of ρ_Q is not calculated with sufficient accuracy by the Savitzky-Golay filter. The error between the filtered value and the plant output is shown in the last column of Fig. 8. The error in $\dot{\rho}_Q$ is almost as large as the signal itself.

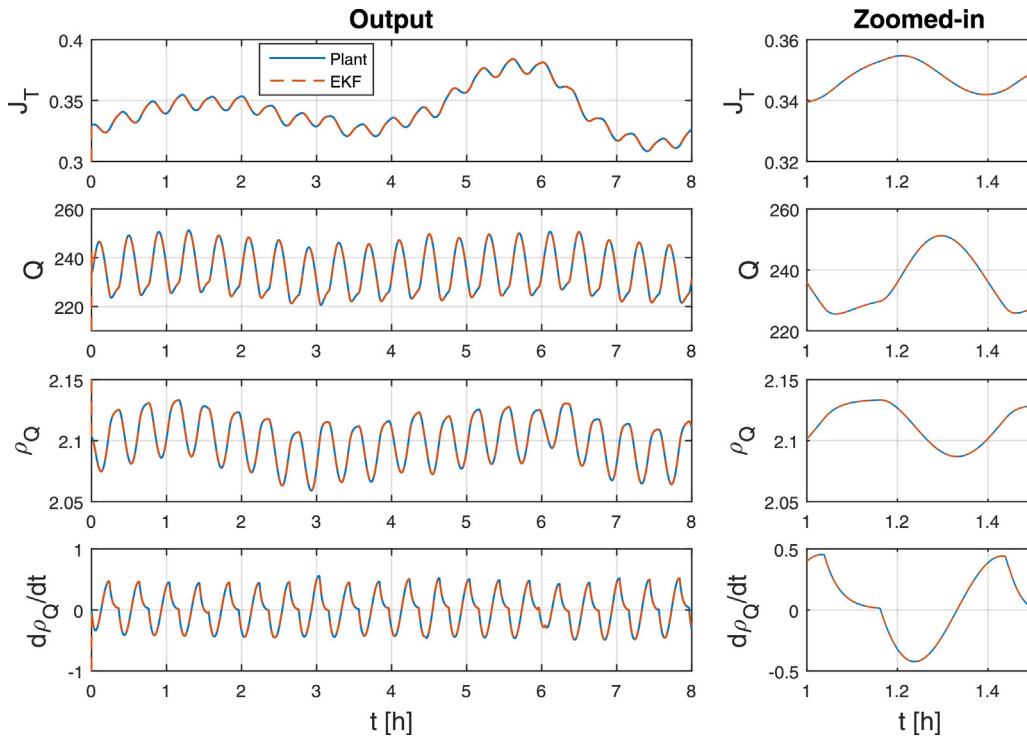


Fig. 6. Output for Simulation Scenario 1.

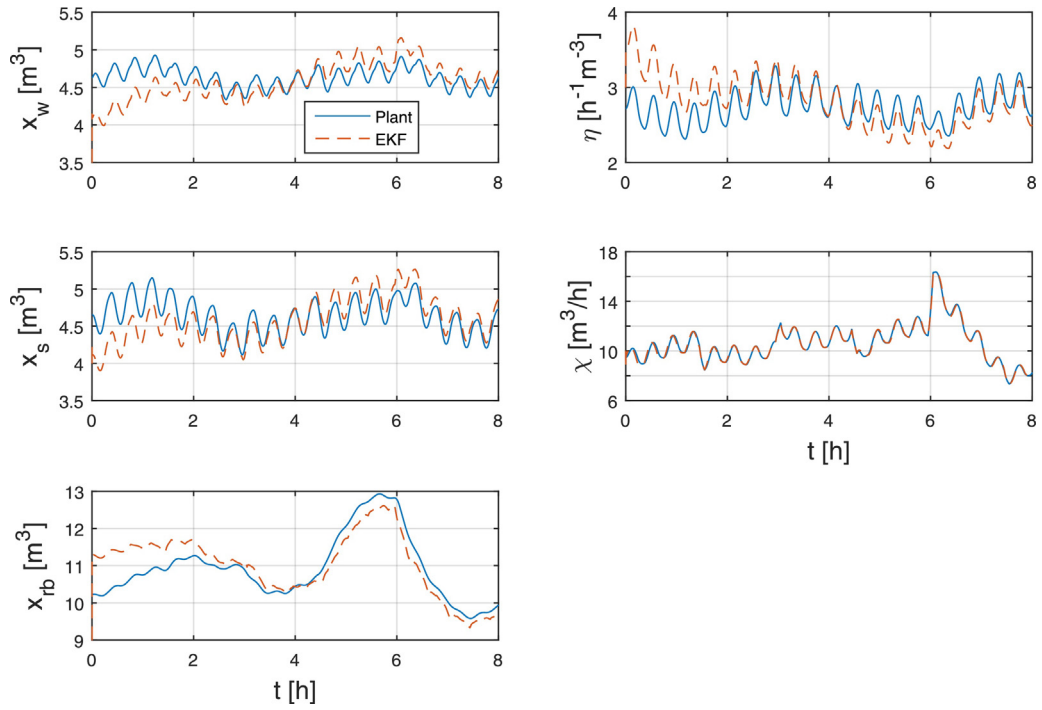


Fig. 7. States and parameters for Simulation Scenario 1.

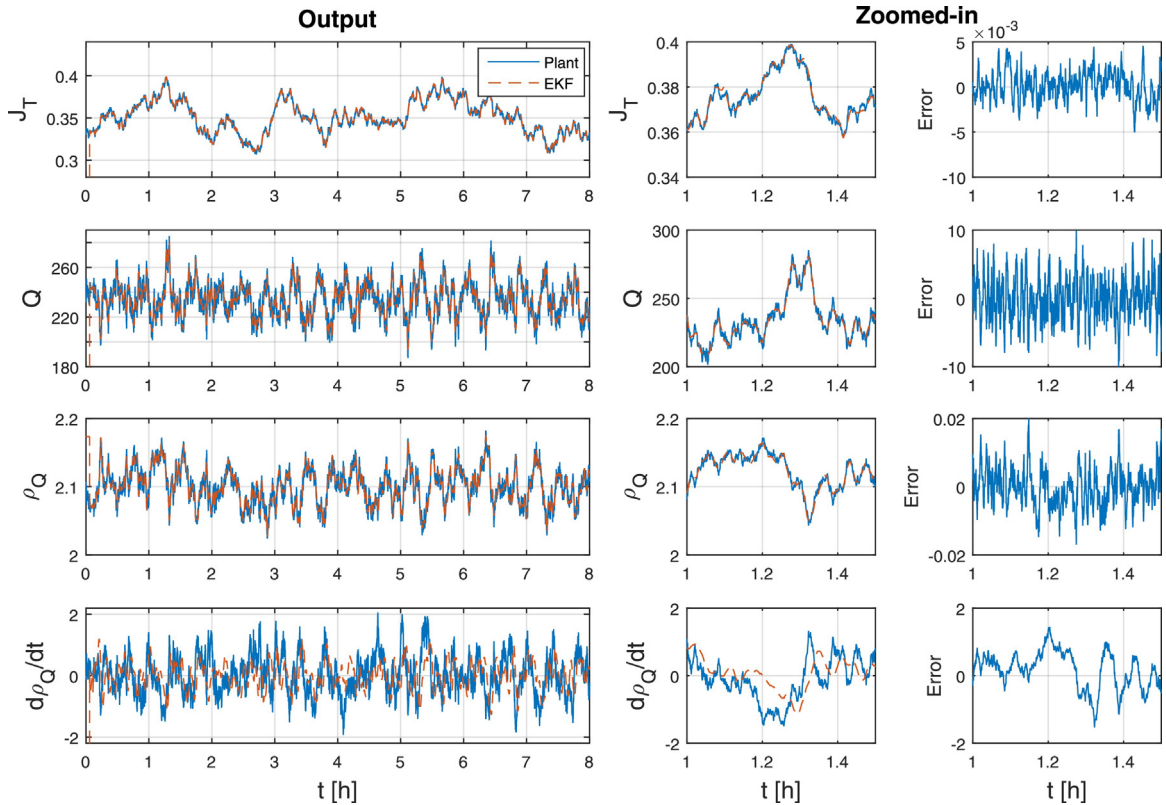


Fig. 8. Output for Simulation Scenario 2. The error shown in the last column is the difference between the EKF output and the plant output.

Since a Savitzky-Golay filter is based on a smoothing least-squares approximation of the signal, the filter cannot guarantee complete noise suppression in high frequencies when calculating the derivative of the signal. The size of the ripples at high frequencies in the magnitude response of the filter can be reduced only if a very long window length is chosen [51]. However, a long window length introduces a time delay (phase shift) which is also undesirable.

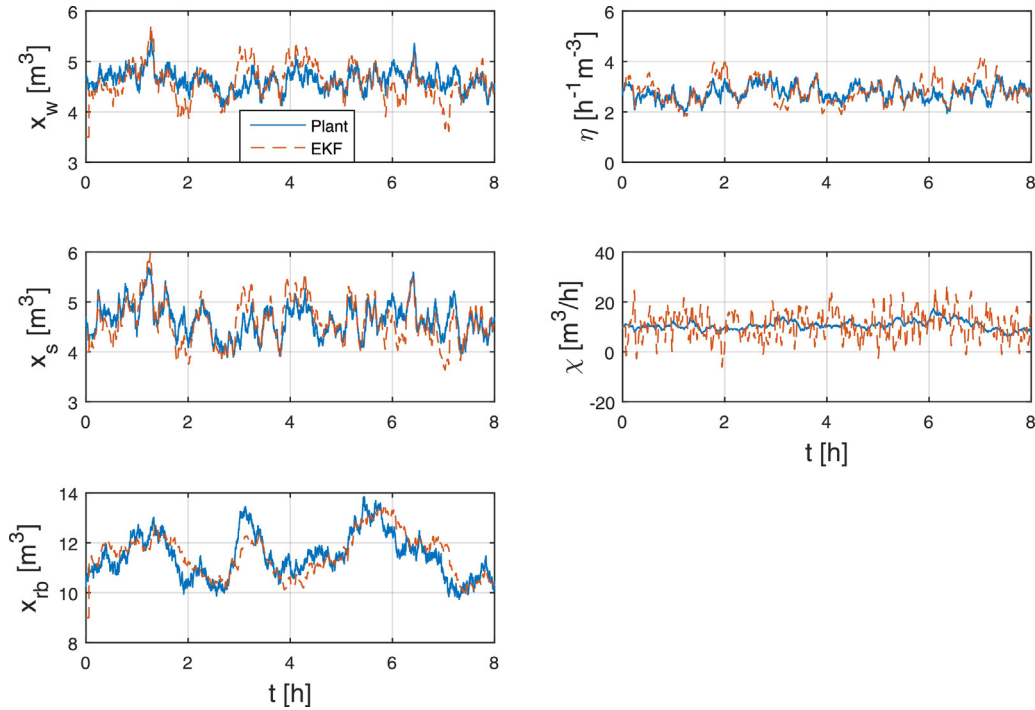


Fig. 9. States and parameters for Simulation Scenario 2.

Table 5

Normalised root mean squared error of state estimation.

	x_w	x_s	x_{rb}	η	χ
NRMSE	6.66%	6.17%	5.14%	12.9%	53.3%

The normalised root mean squared error (NRMSE) between the actual state and the estimated state in Fig. 9 is shown in Table 5 and calculated as

$$\text{NRMSE} = \frac{1}{\bar{\Lambda}} \sqrt{\frac{\sum^N (\Lambda - \hat{\Lambda})^2}{M}}$$

where Λ is the process signal, $\bar{\Lambda}$ is the mean of the process signal, $\hat{\Lambda}$ is the estimate of the process signal, and M is the number of data points. Results in Fig. 9 and Table 5 indicate the EKF filter is able to track x_{rb} with reasonable accuracy, where x_w and x_s are estimated with lower accuracy. Parameters χ and η are used to absorb the uncertainties in the measured data.

8. Conclusion

The states and parameters of a non-linear grinding mill observer model developed in this paper are weakly non-linearly observable, although they are not observable in the linearised case. For linear observability, a reduced observer model was developed. This reduced model represents the constituents of the mill using three states: water, solids, and grinding media (sum of rocks and balls). The grinding environment is modelled using two parameters: a discharge rate and an accumulation rate of solids. The measurement used are: the mill filling volume, the discharge rate of the mill, the discharge density of the mill, and the first time derivative of the discharge density.

Milling data is generated from a model different to the observer model. Simulations indicate that with accurate mill discharge measurements it is possible to estimate the system states and parameters using a discrete-time EKF. The main challenge is to accurately calculate the first time derivative of the mill discharge density.

Since industrial mills rarely include measurement instrumentation at the mill discharge, this work aims to motivate accurate discharge measurements in light of the information about mill inventories that can be gained. This would require careful consideration of mill discharge trommel designs to allow sufficient space to install the required instrumentation. Future work involves using the observer to provide state feedback for an advanced process controller for a grinding mill circuit.

Acknowledgement

The first author gratefully appreciates support from the University of Pretoria postgraduate study abroad bursary programme. The second author gratefully acknowledges financial support provided by the Austrian Academy of Sciences in the form of an APART-fellowship at the Automation and Control Institute of Vienna University of Technology. The third author gratefully acknowledges financial support provided by the Austrian Federal Ministry of Science, Research and Economy, and the National Foundation for Research, Technology and

Development. The fourth author would like to acknowledge the support of the National Research Foundation of South Africa (Grant No. 90533).

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